

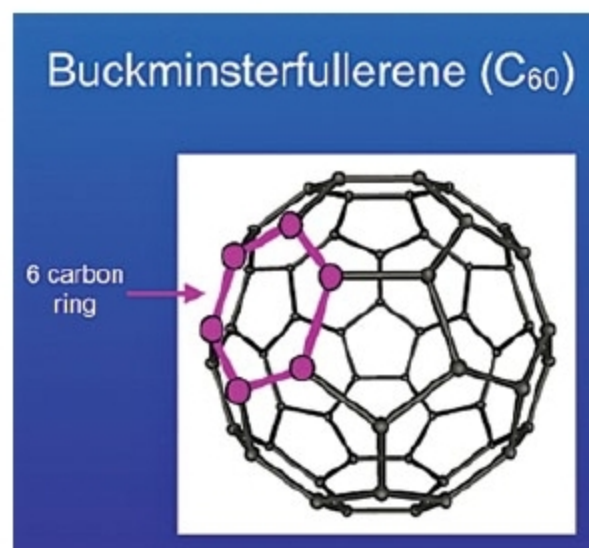


Fig. 3: Model of C₆₀ fullerene

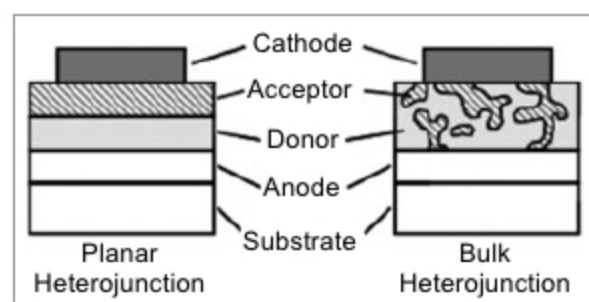


Fig. 4: Basic structure of planar/bulk heterojunction

carbon atoms connected by single and double bonds to form a closed or partially-closed mesh. The molecules may have various shapes like sphere, ellipsoid, tube, etc. These are denoted by their empirical formula written as C_n, where n is the number of carbon atoms. A model of C₆₀ fullerene is shown in Fig. 3.

Once excitons are generated in the absorber material, they need to reach the donor-acceptor interface to be dissociated. Lifetime of excitons is finite and so is their diffusion length, which is around ten nanometres in conjugated molecules. If the donor-acceptor interface is too far from the point where excitons are generated, excitons will decay before reaching the interface and will not yield free charge carriers.

Diffusion of excitons is a severe limit for the performance of donor-acceptor bi-layer. On one hand, thickness of the photoactive layer has to be around hundred nanometres to efficiently absorb light; on the other hand, in a hundred-nanometres-thick bi-layer, only a small fraction of excitons would be generated close enough to the donor-acceptor interface.

One solution to solve this problem is to construct the active layer as BHJ instead of bi-layer, as shown in Fig. 4. Architecture of the BHJ solar cell

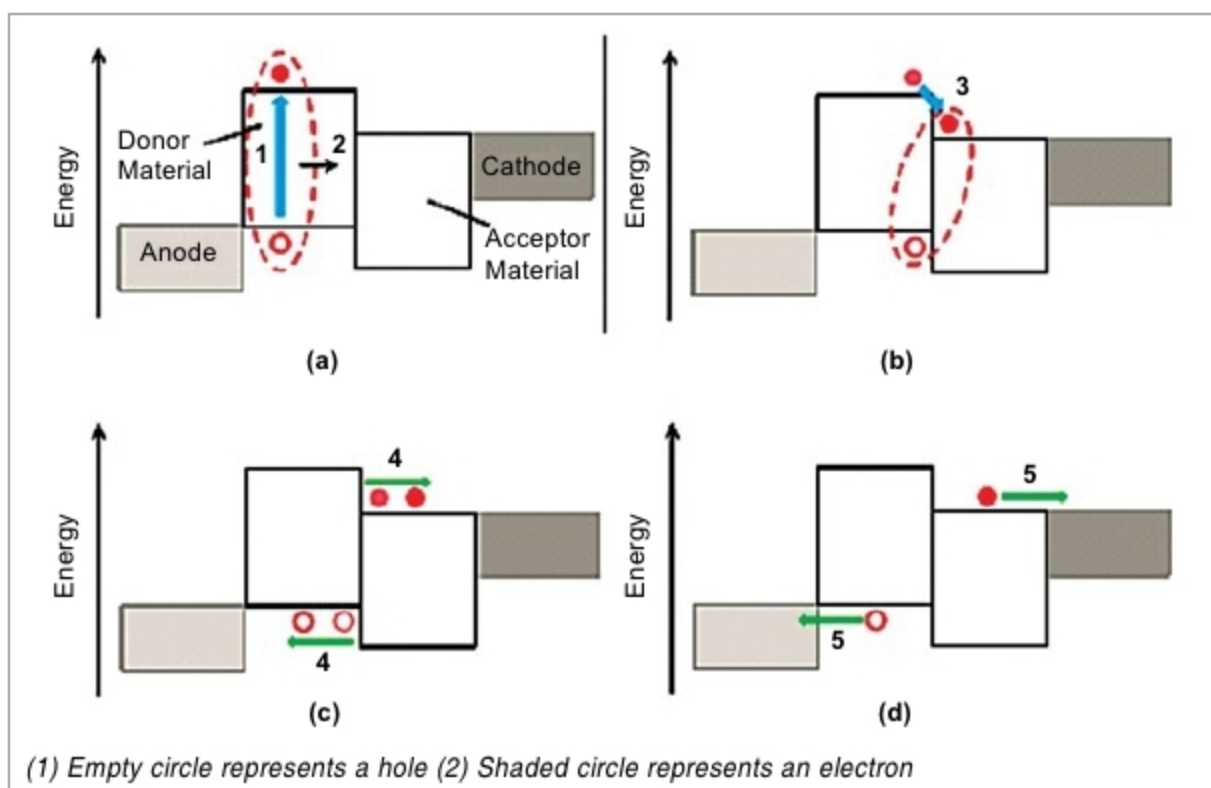


Fig. 5: Operating process of an OPV device

was invented in 1995. BHJ concept involves self-assembly of nano-scale heterojunctions by spontaneous phase separation of donor and acceptor materials.

As a result of self-assembly, charge-separating heterojunctions are formed throughout the bulk of the material. Excitons that diffuse to the donor-acceptor interface can dissociate into free electrons and holes. Connecting the device to an external circuit, free charge carriers are then transported to contacts and extracted from the device, thus generating an electric current.

Negative and positive charges are present in the photovoltaic layer at the same time, thus there is a certain probability of recombination of charges. Recombination represents a loss process for a photovoltaic device because it reduces the number of charge carriers that contribute to photocurrent.

There are three fundamental processes that govern the operation of an OPV device, that is, generation, transportation and recombination of charge carriers, as shown in Fig. 5.

1. Absorption of light generates an exciton marked as 1, which eventually diffuses towards donor-acceptor interface.

2. Excitons are dissociated via charge-transfer, and a bound electron-

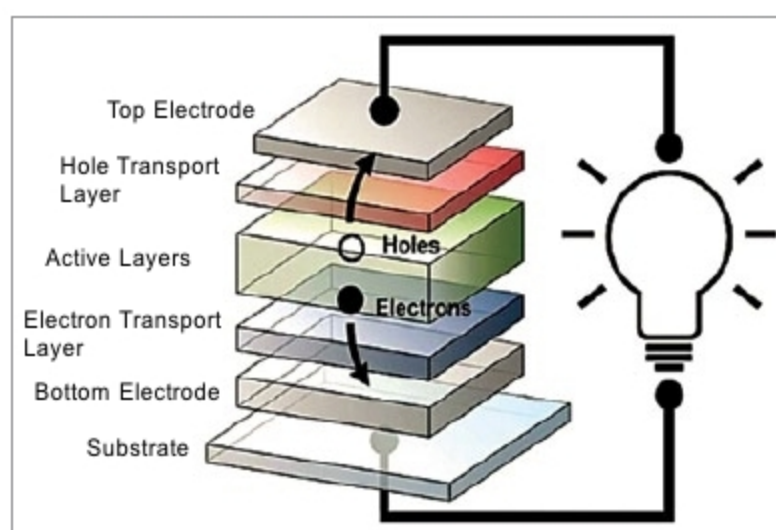


Fig. 6: Structure and operation of an organic solar cell

hole pair (charge transfer state) may be formed.

3. A further dissociation yields free charge carriers.

4. Carriers are transported to electrodes.

Organic solar cell structure and operation

As shown in Fig. 6, when sunlight strikes an organic solar cell, electrons in organic active layers pick up energy and begin moving through the core, leaving behind vacancies, called holes, which essentially move in the opposite direction. Electrons come out of the cell via one electrode, flow along an external circuit, powering a device along the way, and re-enter the solar cell through the second electrode, re-joining the holes they left behind. An electron transport layer and a hole transport layer facilitate the movement of electrons towards one electrode, and holes